

Darshil Kalyani

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EDUCATION

Concordia University

Master of Applied Computer Science

Sept. 2025 – Present

Montreal, QC

Indus University

Bachelor of Technology in Computer Science and Engineering

Sept. 2021 – May 2025

Ahmedabad, India

TECHNICAL SKILLS

Languages: Java, Python, TypeScript, JavaScript, PHP, SQL, HTML, CSS

Frameworks: React, Next.js, Node.js, Express.js, Django, FastAPI, Play Framework

Databases & Tools: MySQL, MongoDB, SQLite, Git, GitHub Actions (CI), Docker, Jira

Testing: JUnit, Mockito, pytest, PHPUnit, Unit & Integration Testing, API Mocking, Debugging

Architecture: Object-Oriented Programming (OOP), MVC, REST APIs, Concurrency, Agile

Authentication: JWT, OAuth 2.0, bcrypt, HTTP-only Cookies, Sessions, Role-Based Access

EXPERIENCE

Physical Research Laboratory (PRL)

Software Development Intern

Jan. 2025 – Apr. 2025

Ahmedabad / Mt. Abu, India

- Developed Python/Tkinter software to automate telescope, CCD camera, and calibration workflows used by astronomers at PRL and Mt. Abu Observatory.
- Implemented thread-based execution and validation of exposure times, filters, frame counts, and observation sequences to support reliable, long-running observation sessions.
- Integrated AstroArt and MaxIm DL through PyAutoGUI; tested hardware interactions incrementally and validated observation workflows during remote and on-site live telescope sessions.

Plumscope Inc.

Frontend Web Development Intern

May 2023 – July 2023

Ahmedabad, India

- Built and deployed responsive About Us and Contact Us pages using HTML, CSS, and JavaScript; added form validation and supported MySQL-backed contact data persistence.
- Resolved layout inconsistencies across desktop, tablet, and mobile; tested forms and navigation in Chrome, Firefox, and Edge, collaborating with a three-member team through review and deployment.

PROJECTS

NotiLytics | *Java, Play Framework, JUnit, Mockito, JaCoCo*

[GitHub](#)

- Built a news analytics application with object-oriented MVC and service layers, isolating API integration, sentiment analysis, and readability calculations through Guice dependency injection.
- Composed non-blocking News API calls with `CompletableFuture/CompletionStage` and maintained isolated session histories for up to 10 searches per user session.
- Wrote automated tests for controllers, services, utilities, and views; mocked external APIs to verify success and failure paths and configured JaCoCo coverage reporting.

DocuMind | *TypeScript, Next.js, React, Express.js, MongoDB*

[GitHub](#)

- Built a full-stack documentation assistant with REST APIs, MongoDB conversation persistence, and Server-Sent Events streaming through a shared interface for four AI providers.
- Implemented JWT access/refresh tokens, Google OAuth, bcrypt hashing, and HTTP-only cookies; protected API routes and validated upload MIME types and a 25 MB size limit before S3 storage.

DriveNow | *PHP, MySQL, JavaScript, PHPUnit, Docker*

[GitHub](#)

- Built a PHP/MySQL car rental platform with customer and admin portals; used transactions, row-level locking, and overlap checks to prevent conflicting bookings during concurrent booking creation.
- Hardened authentication with bcrypt and expiring, single-use password-reset tokens; added CSRF protection, parameterized SQL, and validated image uploads.
- Added PHPUnit unit and MySQL integration tests for authentication, booking rules, and row-lock behavior; configured GitHub Actions to lint PHP and run the suite against MySQL 8.